

Lagrangian Variable Cosmology final-v1.3: Phenomenological Exploration of a Multi-Rung Generalisation of the Localised Modulation in Late-Time Expansion History

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Abstract

This paper is the formal successor to v1.1 and v1.2 in the LVC working-paper series. The previous papers exhibited (v1.1) a half-twist sine-Gordon kink whose squared quarter-angle reproduces a sech^2 modulation centred at $\theta_c = \ln \varphi$ in $\theta = \ln(1+z)$, and (v1.2) a Coleman-Weinberg realisation of the derivative-of-Gaussian profile R_6 . The present paper does not introduce a new lock. It records the phenomenological exploration of a multi-rung generalisation of the v1.1 modulation: writing $T(z) = 1 + \sum_{n=1}^{n_{\max}} A_n \text{sech}^2((\theta - \theta_{c,n})/w)$ with $\theta_{c,n} = n \ln \varphi$ and $w = (\ln \varphi)^3$, the amplitudes A_n and sign pattern are tested against the same data sets that constrain v1.1. Three observations are recorded. First, an alternating-sign φ -amplified ladder ($A_n = (-1)^{n+1} \varphi^{n-1} A_1$) is excluded by the BAO likelihood at $\Delta\text{BIC} = +9.75$ (PP-excluded, $N = 37$) and $\Delta\text{BIC} = +5.44$ (PP-included, $N = 1738$); the disagreement is at the 6σ level in the PP-included fit. Second, a free fit of the second-rung amplitude A_2 with A_1 locked to its v1.1 value returns $A_2 = +0.0295 \pm 0.020$ on PP-included data; the indication is marginal (1.48σ away from $A_2 = 0$) and the 1σ interval is $[+0.0095, +0.0500]$. Third, the ratio $A_2/A_1 \approx 0.55$ is compatible with several natural-constant candidates ($1/\varphi$, $1/\sqrt{\pi}$, $e^{-1/\varphi}$) within $\Delta\chi^2 < 0.02$; the data do not identify A_2/A_1 uniquely. A separate analysis records that the late-time multiplicative modulation $T(z)$, on its own, cannot recover the Planck- ΛCDM value $\sigma_8 \approx 0.811$ while remaining compatible with the BAO data set; the data-compatible ladders push σ_8 *further* from the Planck value than v1.1 single-rung alone. A non-minimal coupling of v1.1's sine-Gordon scalar to the Ricci scalar, $f(\vartheta) = \exp[-\alpha(1 - \cos(\vartheta/2))]$, recovers $\sigma_8 \approx 0.811$ at $\alpha \approx 0.23$ (v1.1 single) or $\alpha \approx 0.12$ (two-rung ladder); the value of α is treated as a phenomenological EFT parameter and not assigned a natural-constant identification. The v1.1 lock pattern is unaffected by the present paper.

1 Introduction

The phenomenological status of a localised multiplicative modulation $T(z) = H(z)/H_{\Lambda\text{CDM}}(z)$ in the late-time expansion rate is documented in v1.1 [1] and v1.2 [2] of this working-paper series. v1.1 records the kink-sector reconstruction of a sech^2 profile in $\theta = \ln(1+z)$ as the half-twist sine-Gordon kink, with locked values $\theta_c = \ln \varphi$, $w = (\ln \varphi)^3$, $A_1 = (\ln \varphi)^4$, and $\Omega_m = 1/(1 + e^{3 \ln \varphi / \varphi})$. On the PP-excluded data set ($N = 37$) the v1.1 full-lock construction returns $\Delta\text{BIC} = -26.44$ against ΛCDM ; on PP-included data ($N = 1738$) the value is -30.31 . v1.2 records an alternative Lagrangian reconstruction for the R_6 derivative-of-Gaussian profile under the same organising principle.

The present paper records a phenomenological question that is left open in v1.1: if the data prefer a localised sech^2 feature at $\theta_c = \ln \varphi$, is this feature the entire modulation, or is it the first of several rungs at $\theta_{c,n} = n \ln \varphi$ for integer n ? The structural motivation is the self-similar lattice $\{n \ln \varphi\}_{n \geq 1}$ in θ -space; no claim of microscopic origin is attached.

The paper is organised in four parts. Section 2 states the multi-rung ansatz. Section 3 records its fit to the v1.1 data sets. Section 4 records the consequences for the linear matter perturbation amplitude σ_8 . Section 5 records a Lagrangian-level extension that resolves the σ_8 behaviour at the cost of one phenomenological coupling. Limitations are stated in Section 6.

The discipline of the paper is the same as v1.1 and v1.2: phenomenology and reconstruction are kept separate, and no claim is attached to the ansatz beyond the consistency of its fits with the data. The v1.1 lock pattern is taken as input throughout and is not modified.

2 Multi-rung ansatz

Writing $\theta = \ln(1 + z)$, the ansatz is

$$T(z) = 1 + \sum_{n=1}^{n_{\max}} A_n \operatorname{sech}^2\left(\frac{\theta - \theta_{c,n}}{w}\right), \quad \theta_{c,n} = n \ln \varphi, \quad w = (\ln \varphi)^3. \quad (1)$$

The first rung is fixed to v1.1: $A_1 = (\ln \varphi)^4 \approx 0.05362$. The remaining amplitudes $\{A_n\}_{n \geq 2}$ and the truncation $n_{\max}$ are inputs to be tested. The Ω_m lock and the width w are unchanged from v1.1. The positions $\theta_{c,n}$ are the $\{n \ln \varphi\}$ lattice; no internal principle of the construction selects this lattice over alternatives, and the question of whether the data prefer it is part of what the paper records.

Two specific one-parameter families are tested:

- *Alternating-sign, geometric amplification*: $A_n = (-1)^{n+1} \varphi^{n-1} A_1$. One free parameter is the amplification ratio (set to φ); the sign pattern is alternating.
- *Same-sign, geometric damping*: $A_n = \varphi^{-(n-1)} A_1$. One free parameter is the damping ratio (set to $1/\varphi$); the sign pattern is uniform.

For the free-amplitude case the second-rung amplitude A_2 is fitted to the data with A_1 locked.

3 Fit to the v1.1 data sets

3.1 Pipeline

The likelihood pipeline is identical to v1.1: DESI DR1 BAO [3], DESI DR2 BAO [3], BOSS DR12 [4], eBOSS DR16 [5], Planck 2018 compressed distance priors [6], SH0ES H_0 [7] (PP-excluded, $N = 37$); and the same with Pantheon+ unbinned supernovae [8] added (PP-included, $N = 1738$). The recombination r_d is derived geometrically from $\theta_* \cdot D_M(z_*)$ as in v1.1. With the v1.1 locks (θ_c , w , A_1 , Ω_m) in place and the ansatz (1) substituted, only $(H_0, \Omega_b h^2)$ and the test parameters of the $\{A_n\}_{n \geq 2}$ sector are fitted.

The v1.1 single-rung reproduction on the present pipeline returns $\chi^2 = 78.962$ for Λ CDM ($k = 3$, PP-excluded) and $\chi^2 = 56.134$ for the v1.1 lock ($k = 2$), recovering the v1.1 paper values to within 10^{-3} . On PP-included the Λ CDM value is 1624.848 and the v1.1 lock returns 1601.999, recovering $\Delta\text{BIC} = -30.31$ to four decimal places.

3.2 Family fits

Table 1 records the fits of the two one-parameter ladder families at $n_{\max} = 2$ and $n_{\max} = 3$ against the PP-excluded and PP-included baselines.

3.3 Free second-rung amplitude

Table 2 records the profile likelihood of the second-rung amplitude A_2 with A_1 locked to its v1.1 value, $\theta_{c,2} = 2 \ln \varphi$, and $w = (\ln \varphi)^3$. The profile is on PP-included data ($N = 1738$).

Three observations follow.

Model	k	χ^2 (PP-excl)	ΔBIC	χ^2 (PP-incl)	ΔBIC
ΛCDM (baseline)	3	78.962	0	1624.848	0
v1.1 single rung	2	56.134	-26.44	1601.999	-30.31
Alt. sign, $A_n = (-1)^{n+1}\varphi^{n-1}A_1$, $n_{\text{max}} = 2$	2	92.322	+9.75	1637.747	+5.44
Same sign, $A_n = \varphi^{-(n-1)}A_1$, $n_{\text{max}} = 2$	2	53.719	-28.85	1599.837	-32.47
Same sign, $A_n = \varphi^{-(n-1)}A_1$, $n_{\text{max}} = 3$	2	53.282	-29.29	1599.450	-32.86

Table 1: Combined fits, ΔBIC against ΛCDM on the same data set. The alternating-sign φ -amplified ladder is excluded by the BAO likelihood; the same-sign φ -damped ladder is of comparable strength to v1.1 single-rung at a small improvement.

Configuration	A_2	χ^2	$\Delta\chi^2$ vs free best
$A_2 = 0$ (v1.1 single rung)	0	1601.999	+2.19
A_2 free (3-parameter fit)	+0.0295	1599.805	0
$A_2 = +(\ln \varphi)^4/\varphi = +0.0331$ (damping lock)	+0.0331	1599.837	+0.03
$A_2 = -(\ln \varphi)^4\varphi = -0.0868$ (amplification lock)	-0.0868	1637.747	+35.75

Table 2: A_2 profile on PP-included data ($N = 1738$). The free fit returns $A_2 = +0.0295$. The 1σ interval ($\Delta\chi^2 < 1$) is $[+0.0095, +0.0500]$. The 2σ interval ($\Delta\chi^2 < 4$) is $[-0.0101, +0.0699]$.

Observation 1: alternating-sign φ -amplified ladder is excluded. The $\Delta\chi^2 = +35.75$ against the free A_2 best is equivalent to a 5.98σ exclusion. On PP-excluded data the exclusion is $\Delta\text{BIC} = +9.75$ relative to ΛCDM .

Observation 2: $A_2 = 0$ is marginally disfavoured. The single-rung case ($A_2 = 0$) sits at $\Delta\chi^2 = +2.19$ against the free best, equivalent to 1.48σ . The result is marginal: the 2σ interval includes zero, and the indication does not strengthen significantly between PP-excluded (1.55σ) and PP-included (1.48σ) fits, suggesting that the Pantheon+ supernovae do not contribute new information on the A_2 amplitude.

Observation 3: A_2/A_1 ratio not uniquely identified. The free fit returns $A_2/A_1 = 0.550$ on PP-included, 0.582 on PP-excluded. A scan of natural-constant candidates within the 1σ range (PP-excluded) returns the following $\Delta\chi^2$ values relative to the in-grid minimum: $1/\sqrt{\pi} = 0.564$: -0.007 ; 0.55 : -0.002 ; 0.60 : -0.007 ; $1/\varphi = 0.618$: 0 ; $e^{-1/\varphi} = 0.539$: $+0.004$. The five candidates lie within $\Delta\chi^2 < 0.02$ of one another; the data do not select between them at present precision.

3.4 Width and position

The width $w = (\ln \varphi)^3$ and the position $\theta_c = \ln \varphi$ of the first rung are unchanged from v1.1; their data preference is documented there. A width scan at $n_{\text{max}} = 1$ on the present pipeline returns the minimum at the v1.1 lock to within 5% of the 1σ interval. The position of the second rung is fixed by the ansatz to $\theta_{c,2} = 2 \ln \varphi$; the data do not constrain this position independently in the present analysis.

4 Implications for σ_8

The linear matter perturbation amplitude $\sigma_8(z=0)$ is computed directly from the growth ODE

$$D'' + \left(2 + \frac{d \ln H}{d \ln a}\right) D' = \frac{3}{2} \Omega_m(a) D \quad (2)$$

(primes are $d/d \ln a$), with $H(z) = H_{\Lambda\text{CDM}}(z) \cdot T(z)$ and matter-era initial conditions $D(a_i) = a_i$, $D'(a_i) = a_i$ at $a_i = 10^{-3}$. The IC is independent of a_i across 10^{-2} to 10^{-5} (Section 6). $\sigma_8(z=0)$ is obtained by multiplying the Planck- ΛCDM reference value 0.811 by the ratio $D_{\text{model}}(a=1)/D_{\Lambda\text{CDM}}(a=1)$.

The same quantity is also computed with CAMB [9] via the **DarkEnergyPPF** module, reverse-engineering $T(z)$ into the effective dark-energy equation of state $w_{\text{DE,eff}}(a)$ and injecting it into the background. The two methods agree to within 0.5% in σ_8 across all models in Table 3, indicating that the dark-energy perturbations as treated in PPF contribute negligibly to σ_8 at the present level.

Model	σ_8 (direct ODE)	σ_8 (CAMB PPF)
ΛCDM at $\Omega_m = \Omega_m^{\text{lock}} = 0.2907$	0.787	0.793
v1.1 single rung	0.788	0.785
Same-sign damped ladder, $n_{\text{max}} = 2$	0.782	0.778
Same-sign damped ladder, $n_{\text{max}} = 3$	0.778	0.774
Alt.-sign amplified ladder, $n_{\text{max}} = 2$	0.806	0.798
Planck- ΛCDM reference	—	0.811

Table 3: $\sigma_8(z=0)$ from background modulation alone. The Planck- ΛCDM reference is the anchor.

Two observations follow.

Observation 4: the data-compatible ladders push σ_8 further from Planck. The same-sign damped ladder, which is preferred by the BAO data (Table 1), returns $\sigma_8 \approx 0.78$, lower than the v1.1 single-rung value and further from the Planck reference 0.811. The model that recovers $\sigma_8 \approx 0.81$ on background modulation alone is the alternating-sign amplified ladder, which is excluded at 6σ by the BAO likelihood.

Observation 5: σ_8 tension is not resolved by late-time multiplicative modulation. Across all $T(z)$ shapes compatible with the BAO data, the predicted σ_8 remains in the range 0.778–0.788, 3.5σ – 5σ below the Planck- ΛCDM value 0.811 ± 0.006 . The mechanism by which the alternating-sign amplified ladder appears to recover $\sigma_8 \approx 0.806$ is the suppression of growth-equation phantom-dark-energy contributions in a redshift range where the BAO data simultaneously constrain $H(z)$; the two constraints are incompatible.

5 Lagrangian-level extension: non-minimal coupling

The observations of Section 4 record that the multiplicative late-time modulation $T(z)$ alone cannot reconcile the BAO and the Planck- σ_8 data sets within the present ansatz. The v1.1 sine-Gordon kink action admits a natural one-parameter extension in which ϑ couples non-minimally to the Ricci scalar:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{2\kappa} f(\vartheta) + \mathcal{L}_{\text{matter}} - \frac{1}{2}(\partial\vartheta)^2 - V(\vartheta) \right], \quad V(\vartheta) = m^2[1 - \cos(\vartheta/2)]. \quad (3)$$

A coupling function of the structurally natural form

$$f(\vartheta) = \exp[-\alpha(1 - \cos(\vartheta/2))] \quad (4)$$

shares the $1 - \cos(\vartheta/2)$ structure with the potential $V(\vartheta)$. The function is strictly positive on all ϑ and generates an effective Newton constant $G_{\text{eff}}(z) = G/f(\vartheta(z))$ on the v1.1 kink trajectory $\vartheta(\theta) = 8 \arctan[\exp((\theta - \theta_c)/w)]$.

The growth equation (2) is modified to

$$D'' + \left(2 + \frac{d \ln H}{d \ln a}\right) D' = \frac{3}{2} \Omega_m(a) \frac{D}{f(\vartheta(a))}. \quad (5)$$

With $H(z)$ kept at its v1.1 (or two-rung ladder) form and α fitted to give $\sigma_8(z=0) = 0.811$, the values returned are:

Background model	α	σ_8
v1.1 single rung	+0.2258	0.811
Same-sign damped ladder, $n_{\max} = 2$	+0.1176	0.811
Same-sign damped ladder, $n_{\max} = 3$	+0.0796	0.811

Table 4: Values of α that recover the Planck-LCDM σ_8 under the coupling (4).

The background impact of $\alpha \neq 0$ on the BAO fit, to linear order in the modified Friedmann equation, is small: the induced shift in $H(z)$ is at the 1% level over $z \in [0.3, 1.0]$ and decreases at higher z . On the PP-excluded pipeline, including this correction in the fit returns a small additional χ^2 improvement ($\Delta\chi^2 \approx -1$). The independent best-fit values of α from background data alone and from the $\sigma_8 = 0.811$ requirement agree in sign and order of magnitude across all three background configurations of Table 4.

The coupling α is treated as a phenomenological EFT parameter. No natural-constant identification is assigned to it in the present paper. The v1.1 lock pattern in the core sector ($\theta_c, w, A_1, \Omega_m$) is preserved without modification.

6 Limitations

Marginal status of the second rung. The free fit $A_2 = +0.0295$ on PP-included data is a 1.48σ indication; the 2σ interval includes $A_2 = 0$. The same-sign damped ladder is compatible with the data and at small ($\Delta\text{BIC} \approx -2$ vs v1.1 single-rung) improvement, but the data do not require the extension.

Non-unique A_2/A_1 identification. The ratio $A_2/A_1 \approx 0.55$ is within 1σ of several natural-constant candidates ($1/\varphi$, $1/\sqrt{\pi}$, $e^{-1/\varphi}$, 0.55, 0.60), with $\Delta\chi^2 < 0.02$ separating them. No unique identification is claimed.

Higher-rung saturation. Truncation at $n_{\max} = 3$ returns $\Delta\text{BIC} = -32.86$ on PP-included, marginally better than $n_{\max} = 2$ at -32.47 . The improvement is below the $\Delta\text{BIC} = 2$ threshold. The data do not require $n_{\max} > 2$.

α is a phenomenological parameter. The coupling α in (4) is fitted to give $\sigma_8 = 0.811$ and is not assigned a natural-constant value. A derivation of α from an underlying ultraviolet theory is not attempted in this paper.

H_0 tension. The combined BAO + distance prior + SH0ES likelihood prefers $H_0 \approx 69$ km/s/Mpc in all configurations tested (v1.1 single rung, same-sign damped ladder, with or without α). The SH0ES local value $H_0 = 73.04 \pm 1.04$ km/s/Mpc remains in tension at $\sim 3.9\sigma$. Resolution of this tension is outside the scope of the late-time modulation framework of the present paper.

Frame dependence and DE perturbations. The non-minimal coupling (4) is treated to linear order in α for the background, and the perturbation analysis uses the G_{eff} modification of the growth equation. A full Jordan-frame analysis with consistent DE perturbations is not performed; the present analysis records that under standard **DarkEnergyPPF** treatment, DE perturbations contribute less than 0.5% to σ_8 for all models tested.

Independent verification. The DR3 BAO release, when available, is expected to halve the BAO uncertainty in the $z \in [1.0, 2.0]$ range that drives the A_2 constraint. At that precision the 1.48σ indication may strengthen or disappear.

7 Outlook

The v1.1 core sector lock ($\theta_c = \ln \varphi$, $w = (\ln \varphi)^3$, $A_1 = (\ln \varphi)^4$, $\Omega_m = 1/(1 + e^{3 \ln \varphi / \varphi})$) remains unchanged. The present paper records that: (i) a multi-rung generalisation at positions $\theta_{c,n} = n \ln \varphi$ is allowed by the data, with a marginal (1.48σ) preference for a positive second rung and uniform sign; (ii) the σ_8 value preferred by Planck- Λ CDM is not recoverable within the multiplicative background-modulation framework alone, regardless of $T(z)$ shape; (iii) a one-parameter non-minimal coupling of the v1.1 sine-Gordon scalar reproduces the Planck- Λ CDM σ_8 value while preserving the v1.1 BAO fit. The coupling is recorded as a phenomenological EFT parameter.

The decisive external input is DR3 BAO; subordinate inputs are the $f\sigma_8(z)$ compilation and the TDCOSMO time-delay distance posteriors, which the present analysis records as compatible with the preferred configuration at current precision but unresolving at the $\sim 1\sigma$ level.

Code and data availability

The reproduction pipeline, an extension of the v1.1 single-file script `lvc_v04_reproduce.py`, is provided as an ancillary file with this submission and will be archived on Zenodo under the MIT licence following acceptance.

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